

Data Communication & Networks(EE2007)

Date: November 5, 2025

Course Instructor(s)

Mr. Khalid Ijaz

Sessional-II Exam

Total Time (Hrs.): 1

Total Marks: 15

Total Questions: 4

Roll No

Section

Student Signature

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Attempt all the questions.

CLO 3: Analyze the design principles involved in data link layer.

Q1: Assume that, in a Stop-and-Wait ARQ system of a data communication network, the bandwidth of the line is 1 Mbps and 1 bit takes 20 ms to make a round trip. If each data frame has a length of 1000 bits, **analyze** the utilization percentage of the link. [2.5 marks]

CLO 3: Analyze the design principles involved in data link layer.

Q2: A Pure ALOHA network transmits 200-bit frames over a 200 kbps shared channel. If the vulnerable time increases by 25%, **analyze** the channel utilization (%) when the system collectively generates 1000 frames per second. [2.5 marks]

CLO 3: Analyze the design principles involved in data link layer.

Q3: A Wireless LAN employs the non-persistent CSMA protocol over a shared channel of 1 Mbps. Each data frame has a length of 1000 bits, and the propagation delay on the medium is 50 μ s. [1.5+1.5+1.5+1.5 marks]

1. **Analyze** the propagation time (a) as a fraction of the transmission time.
2. Using the approximate throughput efficiency expression for non-persistent CSMA:

$$S = \frac{G e^{-aG}}{1 + 2aG}$$

Where G is the offered load (frames per frame-time), **analyze** the throughput (S) when G=0.5.

3. **Analyze** the channel utilization (%) for G=0.5.
4. **Analyze** how increasing the propagation time (larger a) affects system efficiency and collision probability.

CLO 3: Demonstrate the application of network layer protocols.

Q4: Consider the network shown below, where Host H1 (running Process P1) sends a packet to Host H2 (running Process P2) through a set of routers (A, B, C, D, E) inside an ISP's network. **Demonstrate** the routing tables of all those routers, which are involved in communication process. [4 marks]

